

ABOUT	PhD scholar at the Department of Artificial Intelligence (DoAI), IIT Kharagpur, in the RHPI (Receding Horizon and Predictive Intelligence) Lab under the supervision of Dr. Prabhat Kumar Mishra. My research focuses on developing new robust RL algorithms with applications in LLMs and robotics.		
RESEARCH INTERESTS	Machine Learning Optimization Reinforcement Learning State-Space Models		
EDUCATION	PhD, Artificial Intelligence	2025 – Present	
	Indian Institute of Technology Kharagpur	CGPA 9.33/10	
	MSc, Mathematics	2021 – 2023	
	Indian Institute of Technology Kharagpur	CGPA 8.76/10	
	BSc, Mathematics	2018 – 2021	
Ramakrishna Mission Vidyamandira (University of Calcutta)		CGPA 9.15/10	
Class XII (WBCHSE, 2018) 88.6% · Class X (WBBSE, 2016) 94.57%			
Raiganj Coronation High School (H.S.)			
PUBLICATIONS	Ranjan Sarkar, Prabhat K. Mishra. “LQG Solution for POMDP without estimating states: A minimum variance approach.” Accepted at <i>Indian Control Conference (ICC)</i> , 2027.		
EXPERIENCE	Senior Business Analyst, EXL Service · Gurgaon, India	2023 – 2024	
	- Built data pipelines and analytics with SQL and SAS. - Worked on business-as-usual (BAU) operations.		
PROJECTS	Master’s Thesis: Image Compression	2022 – 2023	
	Supervisor: Prof. Swanand Khare, Department of Mathematics, IIT Kharagpur - Implemented tensor CP decomposition in Python. - Applied the Alternating Least Squares (ALS) optimization method to obtain low-rank approximations of RGB image tensors.		
TEACHING ASSISTANT	AI31201 · Reinforcement Learning	Autumn 2026	
	AI60209 · AI for Robot Autonomy	Autumn 2026	
	AI61006 · AI for Cyber Physical Systems	Spring 2026	
	AI60211 · Algorithmic and Mathematical Foundations for AI	Autumn 2025	
SKILLS	Programming & Tools: Python, C, PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, SymPy, CasADi, SQL, SAS, Jupyter, LaTeX		
	Coursework: Machine Learning, Deep Learning, Artificial Intelligence, Convex Optimization, Non-Linear Programming, Linear Algebra, Probability & Statistics, Real Analysis, Advanced Numerical Techniques, AI for Cyber Physical Systems, AI for Robot Autonomy		
AWARDS	- Qualified Joint CSIR-UGC NET (June 2024) with All India Rank 83. - INSPIRE Scholarship (2018–2023), DST, Govt. of India — awarded for top 1% in the WB HS board examination (2018).		
INTERESTS	Photography, Singing, Swimming, Table Tennis		